

5.1 Causes of climate change

Earth's climate has varied over time. The Quaternary Period is the last 2.6 million years of geological time. During this period, Earth's climate has varied from cold **glacial periods**, or **ice ages**, with massive continental ice sheets, to **interglacial periods** with climates similar to or warmer than today. The last glacial period ended 12 000 years ago and we are currently in an interglacial period called the Holocene. Glacial periods last around 90 000 years, whereas interglacial periods last around 10 000 years.

Climate change refers to changes in temperature (both increases and decreases) and weather patterns over a long period of time. **Global warming**, in contrast, refers only to the increase in average global temperature over the past century. Climate change can occur naturally or it can be caused by humans (anthropogenic climate change). It is normal to have greenhouse gases in the atmosphere; indeed, without them, the **natural greenhouse effect** would not occur and Earth would be uninhabitable. However, humans have now released more **greenhouses gases** into the atmosphere than ever before. This has resulted in the **enhanced greenhouse gas effect**, which has led to global warming and climate change (Figure 5.1).

TIP

Do not confuse global warming with climate change. Climate change describes an increase and decrease in global temperatures, whereas global warming describes an increase in global temperature only.

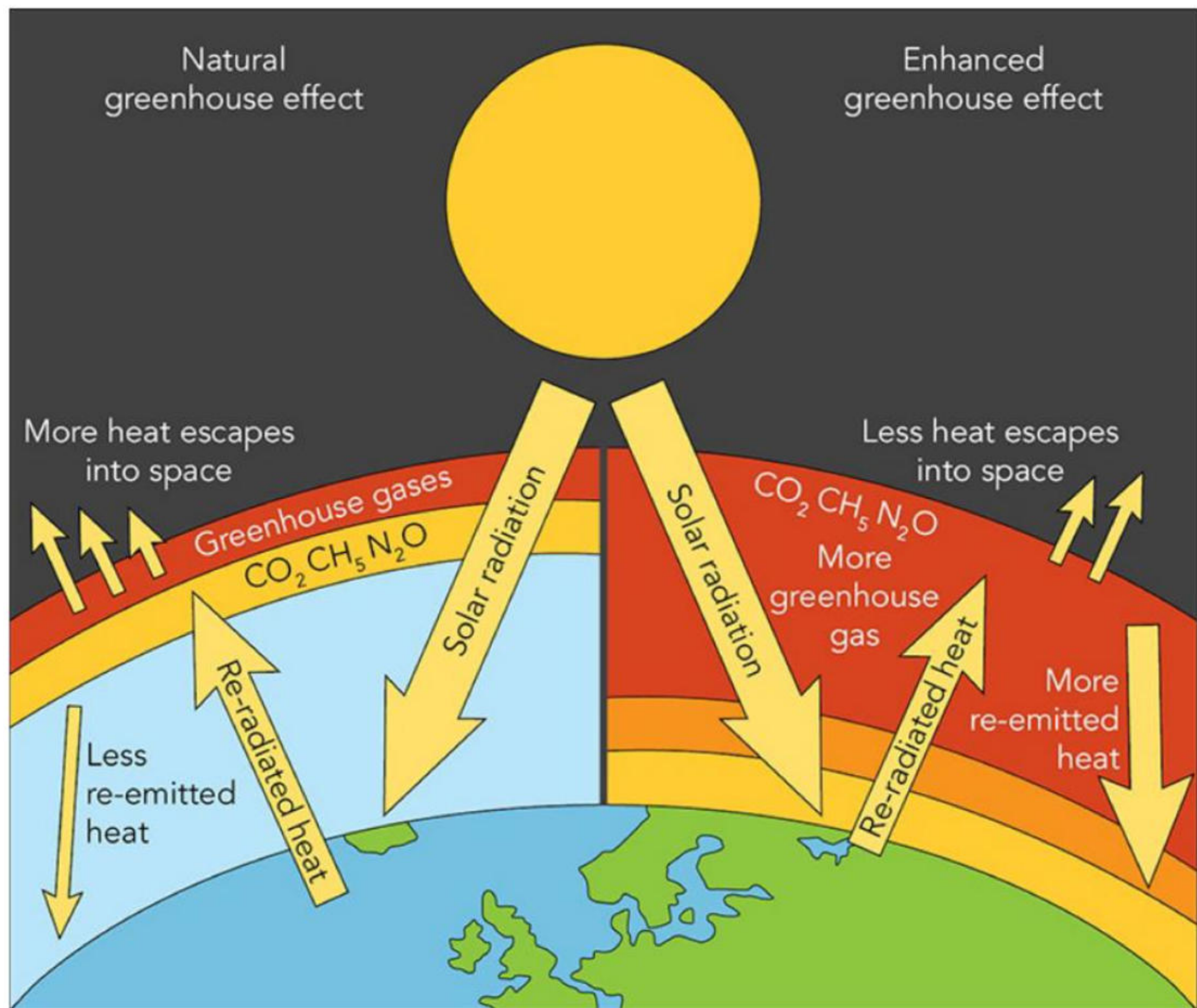
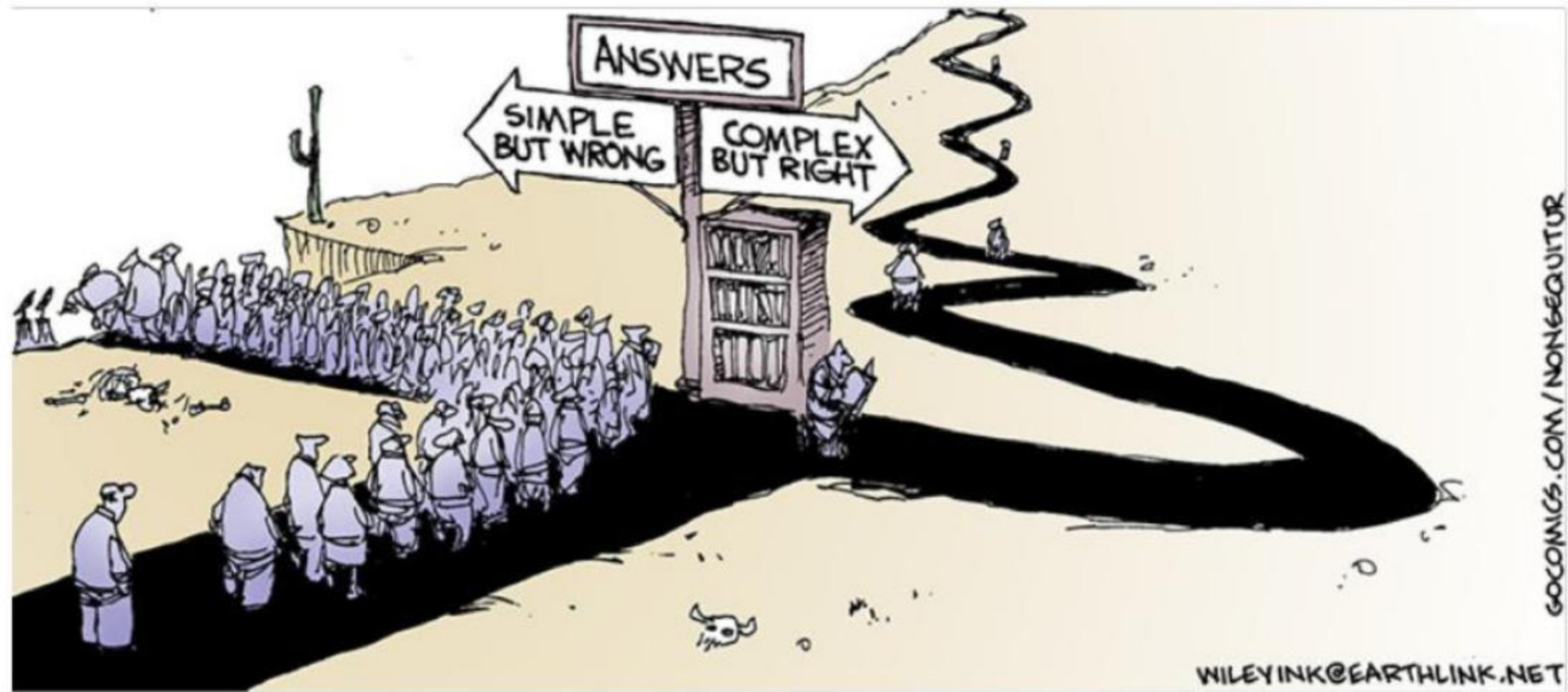


Figure 5.1: The natural greenhouse effect and the enhanced greenhouse effect



Source A: A cartoon about climate change

TASK 1: Study Source A

In pairs, discuss the following questions:

- What do you think the cartoon is trying to say about climate change?
- How do you think this cartoon will be different in 100 years? You could draw your own cartoon to capture your discussion.

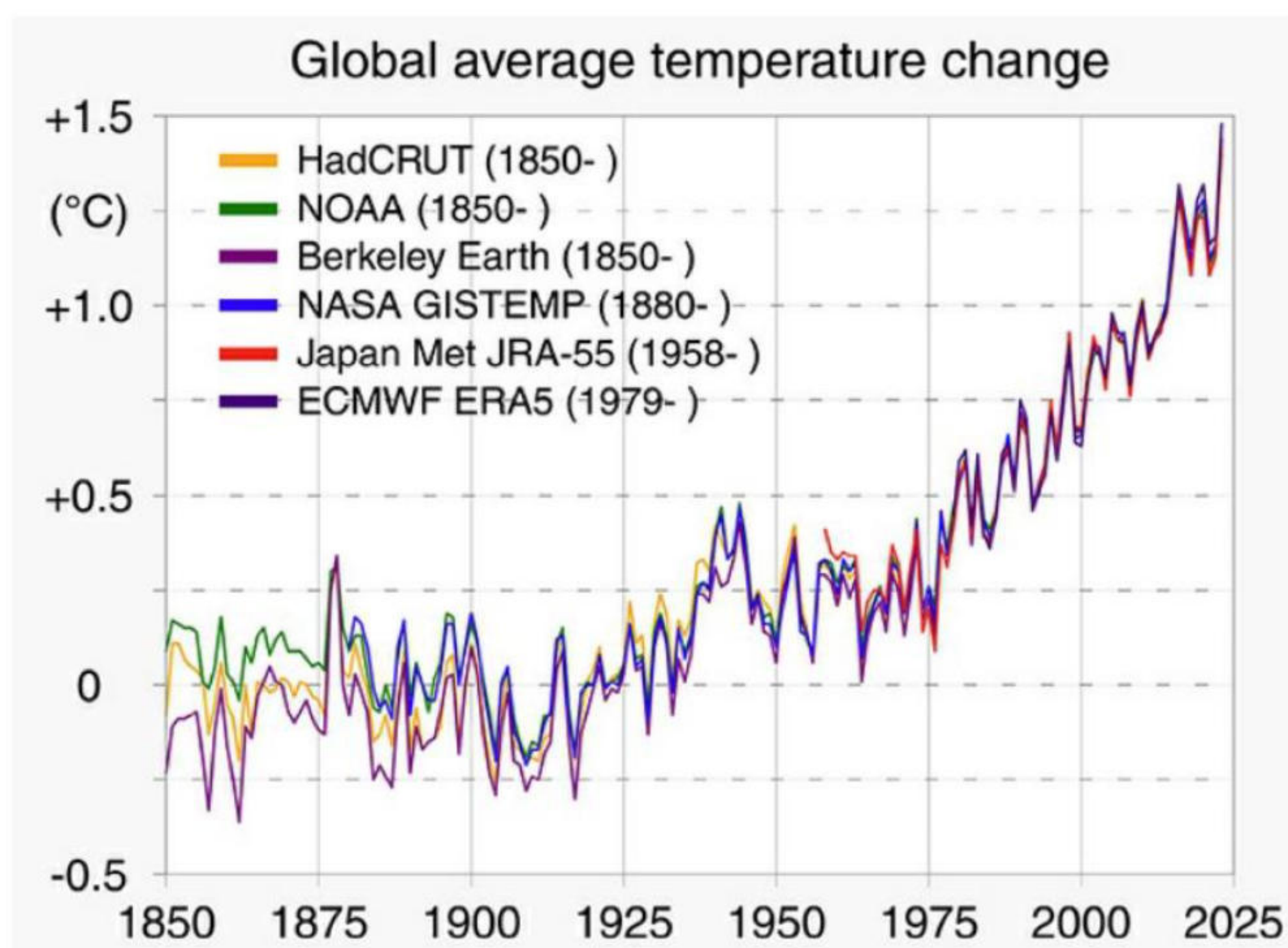
Share your ideas with the class and explain the meaning of your cartoon if you have drawn one.

Evidence for climate change

We know that Earth's climate has changed in the past, as many sources of evidence exist. These include historical global temperature data, ice cores, sea ice positions, as well as historic writings and paintings. Some records go back further in time than others but all are useful to climate scientists.

Global temperature data

Earth's temperature has fluctuated over time (Figure 5.2). However, the oldest temperature records measured by people (direct records) only date back to 1659. This is when scientists in England started measuring the temperature with thermometers every month, before moving to daily recordings from 1772. Today, we measure global temperature constantly by methods such as thermometers in ships, aeroplanes, and buoys at sea.



Writings and paintings

We also know that global temperatures have changed over time due to written records and paintings. For example, we know about the Little Ice Age (a period of cooler temperatures from the 14th to the mid-19th century), from paintings and drawings. Frost fairs, like the one depicted in Figure 5.3, were held on the River Thames in England as early as 1607 when the river froze. Similar frost fairs were commonplace during this time across Europe.



Figure 5.3: The frost fair on the River Thames in London in the winter of 1683–84

Sea ice positions

Global sea ice coverage can be a useful indicator of climate change. Sea ice covers around 34 million square km of Earth (roughly three times the area of Canada). This is equal to around 9% of the world's oceans being covered by sea ice at some point in the year.

Sea ice is prevalent in higher latitudes like the Arctic and the Antarctic for some, or all, of the year. In the Arctic, sea ice coverage has been declining at a substantial rate due to warmer temperatures (losing roughly 13% per decade). In contrast, in the Antarctic, sea ice coverage increased from 1979 (when sea ice records from satellites began) at a rate of around 1.5% per decade until 2016. Recent years, though, have seen a dramatic decrease in Antarctic sea ice extent. Figure 5.4 shows recent sea ice changes for the Arctic and Antarctic.

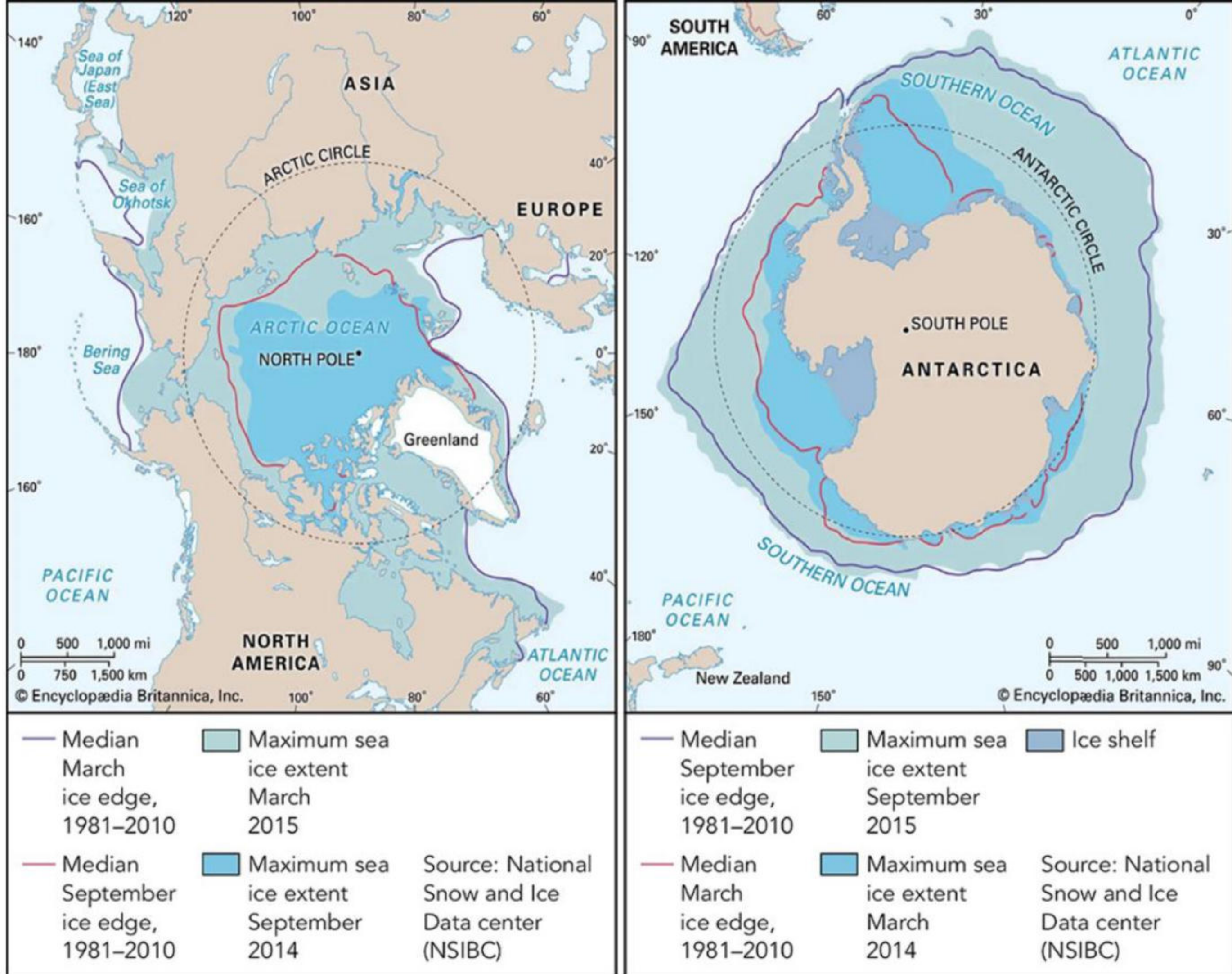


Figure 5.4: Sea ice extent in the Arctic and the Antarctic

Ice cores

To go back further in time, scientists use proxy data (indirect records) to reconstruct past climates. One of the longest proxy records we have comes from ice cores drilled out of ice sheets in Antarctica and Greenland. Scientists can slice up and melt ice cores, which can be several km long. They do this to extract the gas that was frozen in the ice millions of years ago. Scientists can then work out the atmospheric composition of Earth at the time the gas was trapped in the ice. Ice cores from Antarctica have helped climate scientists estimate the global temperature record for the past 800 000 years (Figure 5.5).

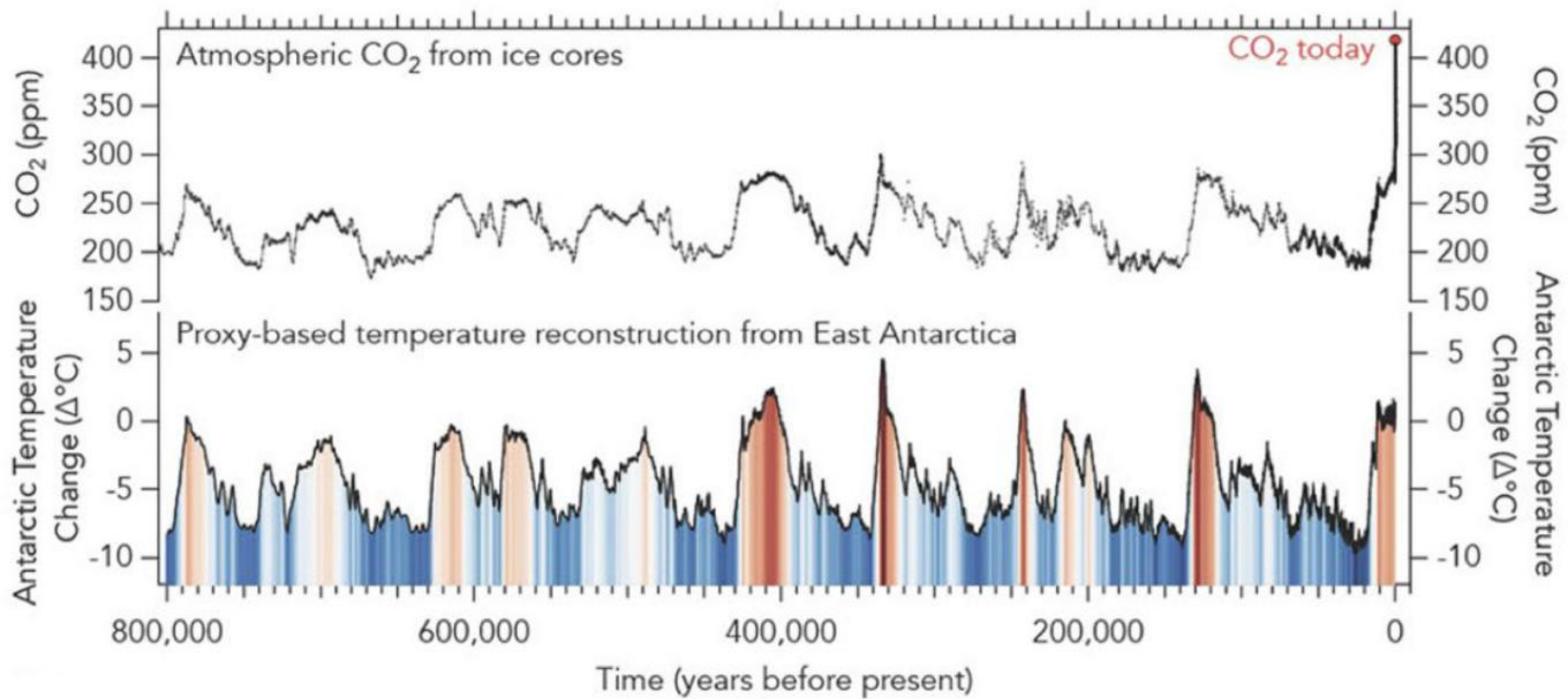


Figure 5.5: Antarctica's temperature record for the last 800 000 years (lower graph). This has been derived from carbon dioxide gas measurements from ice cores (upper graph).

Natural causes of climate change

Milankovitch cycles help explain long-term variations in the Earth's climate over tens of thousands to hundreds of thousands of years. Milankovitch cycles are periodic orbital changes that affect how much solar radiation the Earth receives.

There are three orbital cycles that affect Earth on different time scales (Figure 5.6):

- Changes to Earth's orbit shape (eccentricity) – over around 100 000 years, Earth's orbit changes from an ellipse to a circle due to the gravitational pull of Jupiter and Saturn. A circular orbit has more even temperatures across the seasons than a more elliptical orbit.
- Changes to Earth's 'wobble' (axial precession) – as Earth spins, its axis wobbles in a circle. It takes around 26 000 years to complete a full circle. Axial precession makes seasonal contrasts more extreme in one hemisphere, and less extreme in the other. Currently, summers in the Southern Hemisphere are hotter than the Northern Hemisphere, where the seasonal variations are more moderate. In around 13 000 years, axial precession will cause these conditions to flip.
- Changes to Earth's tilt (obliquity) – over a period of around 41 000 years, the tilt of Earth varies from 22.1–24.5°. It is currently 23.4° and very slowly decreasing. The greater the angle of Earth's axial tilt, the more extreme the seasons are. This is because each hemisphere receives more solar radiation during its summer, when the hemisphere is tilted toward the sun, and less during winter, when it is tilted away.

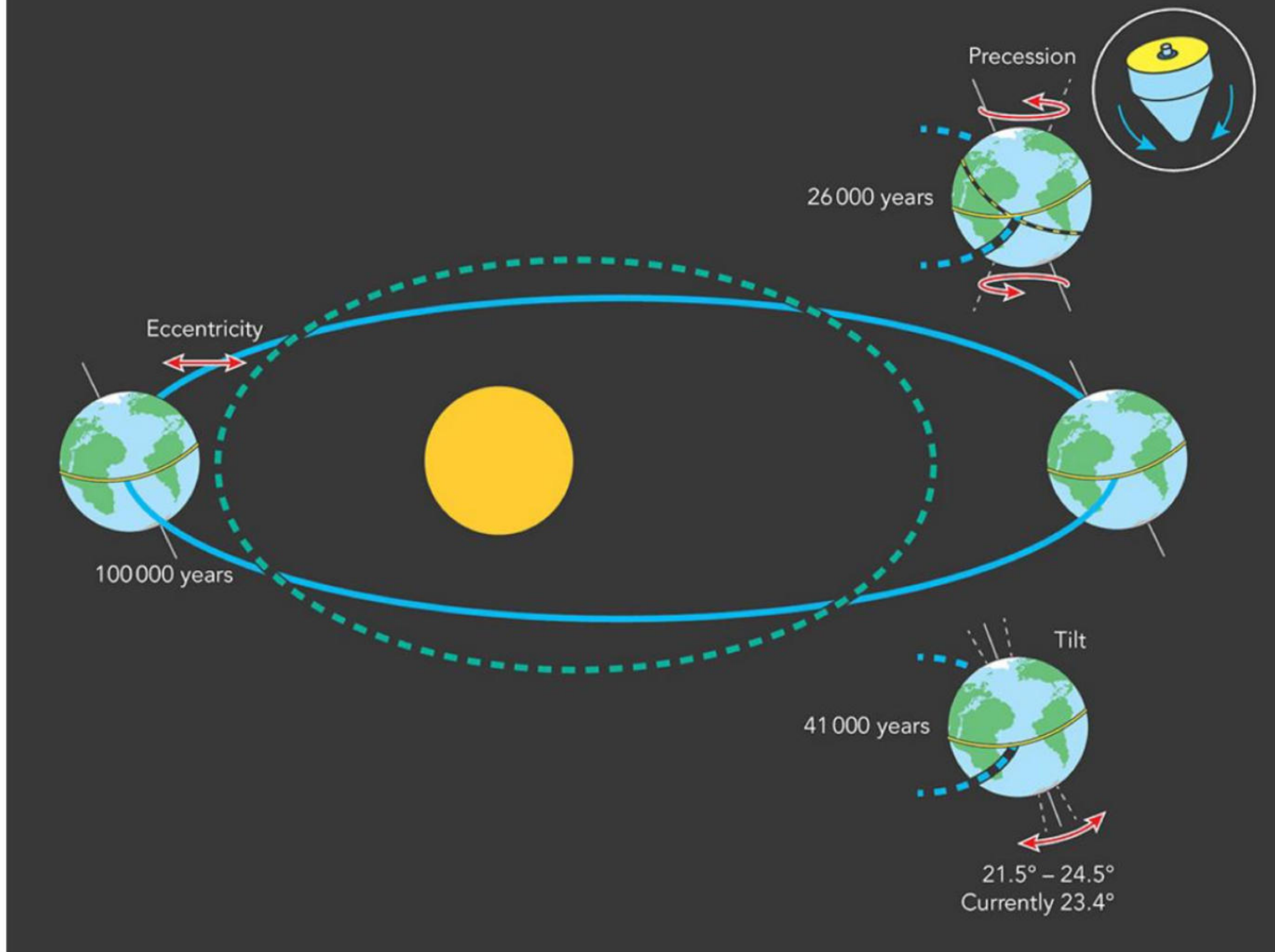


Figure 5.6: Milankovitch cycles

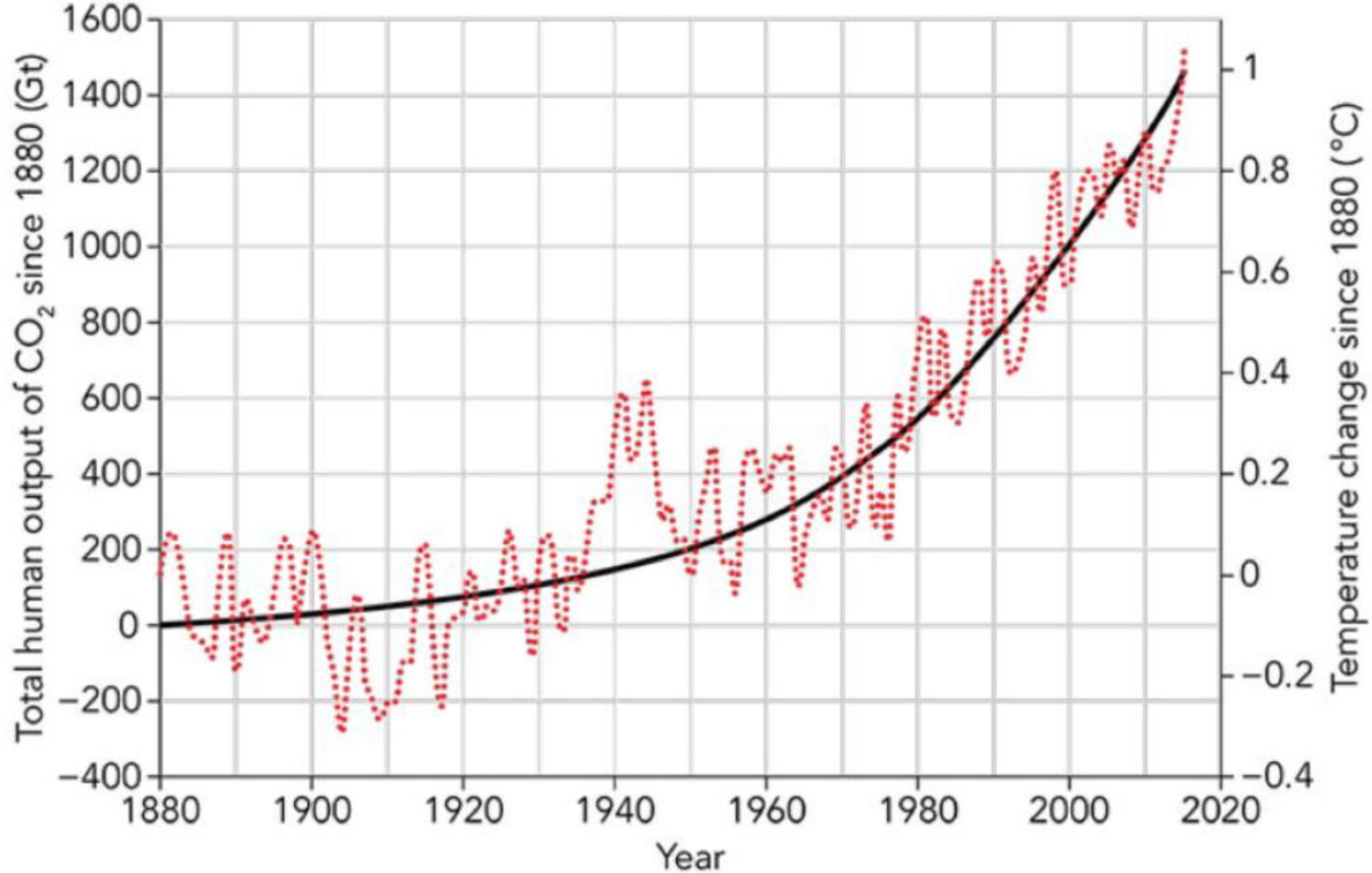
Another natural cause of climate change is sunspot activity. Sunspots are dark areas on the surface of the sun. Historical records suggest that fewer sunspots result in cooler temperatures, whereas more sunspots result in higher temperatures on Earth.

Volcanic activity also affects the global climate. Volcanic eruptions can both cool and warm the planet. Gases like carbon dioxide that a volcano releases during an eruption warm the planet. **Aerosols** like ash and sulphur dioxide that volcanoes emit during an eruption can temporarily cool the global climate as these small particles reflect solar radiation away from Earth. If a supervolcano, like Yellowstone in the USA, were to erupt, Earth could be pushed back into an ice age.

Human causes of climate change

The main cause of climate change by humans is burning **fossil fuels** like coal, oil, and natural gas. This has had a significant impact on global temperatures since the industrial revolution in the 19th century. Global temperatures have risen by 1.2 °C since 1880.

Burning fossil fuels produces greenhouse gases, including carbon dioxide, methane, nitrous oxide, and water vapour. As outlined in Figure 5.1, these gases become trapped in the atmosphere and insulate Earth, similar to a blanket. This causes the planet to heat up. Source C shows the amount of global greenhouse gas emissions emitted by sector.



Source B: Total emissions of carbon dioxide and global temperature change between 1880 and 2018

TASK 2: Study Source B

- Describe the graph.**
- In the year 2000, what was the total human output of carbon dioxide?**
- What is the range** of total human output of carbon dioxide between 1880 and 2018?

TASK 3: Study Source C

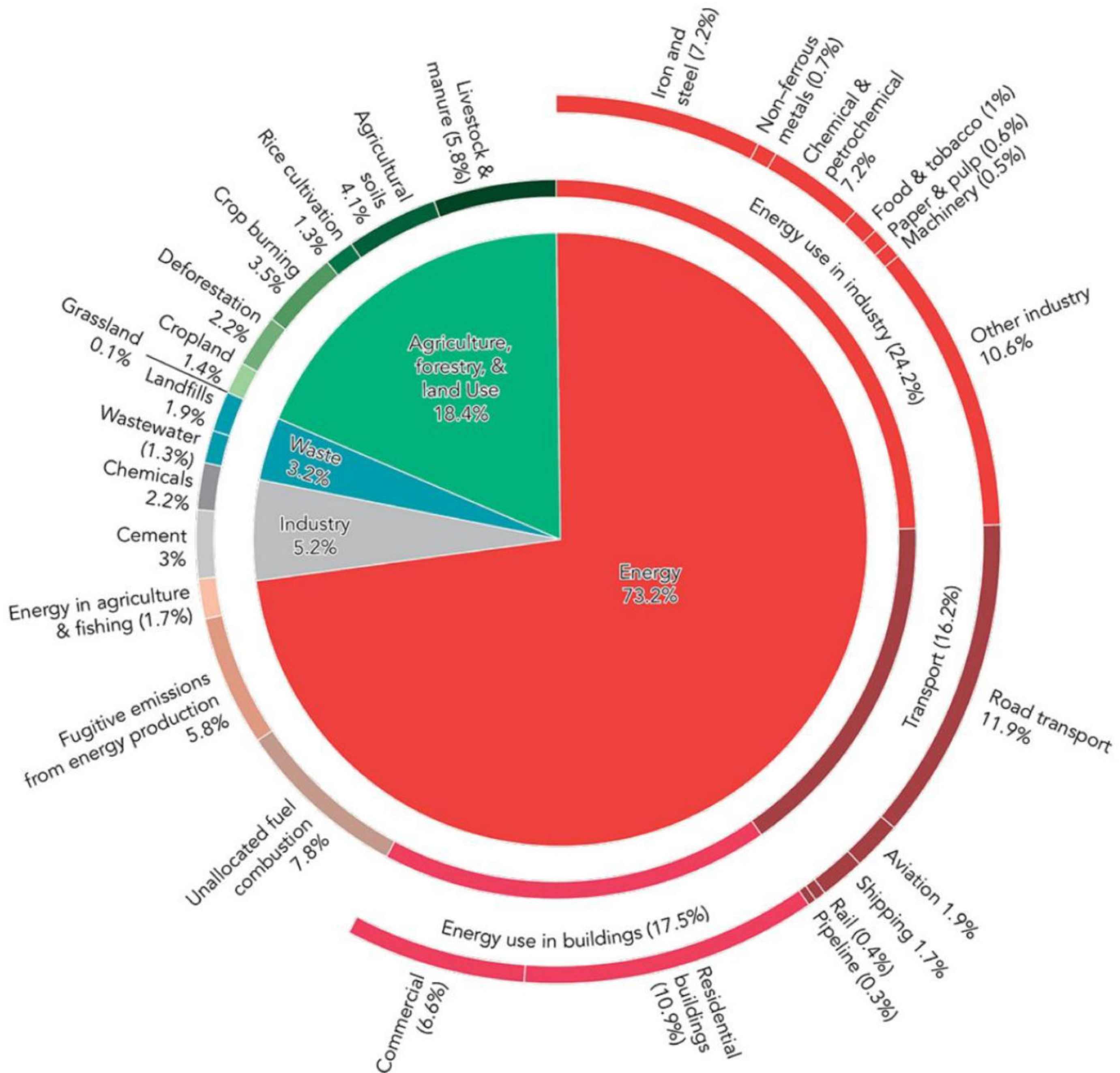
- Using the data in the table below, **construct a pie chart** that shows the greenhouse gas emissions by sector for the USA.

Sector	Greenhouse gas emissions (%)
Energy	63
Agriculture, forestry, and land use	10
Waste	4
Industry	23

- Describe the differences** between Source C and your pie chart.
- In pairs, suggest three possible reasons for these differences.

Global greenhouse gas emissions by sector

This is shown for the year 2016 – global greenhouse gas emissions were 49.4 billion tons GO_2eq



Source : Global greenhouse gas emissions by sector (2016)

► Image Description

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Deforestation also contributes to climate change. Carbon dioxide is released into the atmosphere through burning at the time of deforestation and through the decay of plant material in the following years. This contributes to the greenhouse effect. Trees also act as carbon sinks, as they absorb carbon dioxide during photosynthesis. Fewer trees reduce Earth's capacity to absorb carbon dioxide. According to the World Wildlife Fund, deforestation is responsible for roughly 15% of global greenhouse gas emissions. Deforestation is a particular issue in tropical regions.

Agriculture contributes to climate change in a number of ways:

- Due to an increasing demand for meat, humans are rearing more animals, such as cattle. Cattle produce methane during digestion. This methane is then released into the atmosphere. Methane is a greenhouse gas which has a high **global warming potential (GWP)**.
- Clearing forests to make agricultural land, especially in environments like rainforests, releases carbon and reduces the number of trees available to absorb carbon dioxide through photosynthesis.
- Rice paddies release methane gas during cultivation (Figure 5.7).



Figure 5.7: Rice paddy cultivation is a significant contributor of methane into the atmosphere

Table 5.1: The global warming potential of greenhouse gases

Greenhouse gas	GWP over 100 years
Carbon dioxide (CO ₂)	1
Methane (CH ₄)	29.8
Nitrous oxide (N ₂ O)	273
F-gases	1000–10 000

TASK 4: Study Table 5.1

- Whether the source is natural or human, different greenhouse gas emissions warm the atmosphere differently. This is called the global warming potential (GWP) Fluorinated gases (F-gases), for example, are 10 000 times more efficient at trapping heat than carbon dioxide.
- a** Using Table 5.1 and the information presented earlier in this topic, which greenhouse gas do you think is the most significant cause of global warming? Which is the least significant? Support your ideas with evidence.
 - b** Why is it important to address the emission of all greenhouse gases and not just focus on one particular gas?